Ans. (b) : National Housing Bank was established in July 1988 as a subsidiary of Reserve Bank of India. This bank is the apex bank for housing finance in the country.

58. Which one of the following is the correct sequence in the decreasing order of contribution of difference sectors to the Gross Domestic Product of India?

(a) Services-Industry-Agriculture
(b) Services-Agriculture-Industry
(c) Industry-Services-Agriculture
(d) Industry-Agriculture-Services

Ans. (a) : The contribution of the given sectors in the GDP of India in descending order is as follows - Services, Industry and Agriculture.
According to the Economic Survey 2006-07, the share of service sector in GDP is 55.1%, the share of agriculture sector is 18.5% and the share of industrial sector is 26.4%.
The service sector is the largest sector of India. Gross Value Added (GVA) at current prices for the service sector is estimated at 96.54 lakh crore INR in 2020-21. The service sector accounts for 53.89% of total India's GVA of 179.15 lakh crore Indian rupees. With GVA of Rs. 46.44 lakh crore, the Industry sector contributes 25.92%. While Agriculture and allied sector share 20.19%.

59. Consider the following statements:
1. The repo rate is the rate at which other banks borrow from the Reserve Bank of India.
2. A value of 1 for Gini Coefficient in a country implies that there is perfectly equal income for everyone in its population
Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : Repo Rate is the rate at which commercial banks borrow funds from RBI. A reduction in the repo rate will help banks to get money from the central bank at a cheaper rate. When the repo rate increases borrowing from RBI becomes more expensive. A value of (0) for Gini Coefficient in a country implies that there is perfect equality in the country. If the value is 1 then there is complete inequality of income in the country.

60. Basel II relates to which one of the following?

(a) International standards for safety in civil aviation.
(b) Measures against cyber crimes.
(c) Measures against drug abuse by sportspersons.
(d) International standards for measuring the adequacy of a bank's capital.

Ans. (d) : Basel II is the revised international capital framework. The Basel II framework describes a more comprehensive measure of minimum standard for capital adequacy that national supervisory authorities are now working to implement through domestic rule-making and adoption procedures.

PHYSICS

61. What is the average distance (approximate) between the Sun and the Earth?

(a) 70×10^5 km (b) 100×10^5 km
(c) 110×10^6 km (d) 150×10^6 km

Ans. (d) : The average distance between the sun and the earth is 152×10^6 km which is near to 150×10^6 km.

62. Consider the following statements:
1. If magenta and yellow coloured circles intersect, the intersected area will have red colour.
2. If cyan and magenta coloured circles intersect, the intersected area will have blue colour.
Which one of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : The intersected area of magenta and yellow-coloured circles will have red colour. The intersected area of cyan and magenta coloured circles will have blue colour.
Magenta = Red + Blue
Cyan = Blue + Green

63. Consider the following statements:
1. A flute of smaller length produces waves of lower frequency.
2. Sound travels in rocks in the form of longitudinal elastic waves only.
Which one of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b) : 1. Frequency produced by a string is inversely proportional to its length.
$f=2Lv$
2. Sound always travels as longitudinal wave through the propagating material.
So only statement 2 is correct.

64. Four wires of same material and of dimensions as under are stretched by a load of same magnitude separately. Which one of them will be elongated maximum?

(a) Wire of 1 m length and 2 mm diameter.
(b) Wire of 2 m length and 2 mm diameter.
(c) Wire of 3 m length and 1.5 mm diameter.
(d) Wire of 1 m length and 1 mm diameter.

Ans. (c) : As per Hookes law

$\Delta L \propto \frac{L}{D^2}$, So ΔL for 1m length and 2 mm dia –

$\frac{1000}{4} = 250\text{mm}$

ΔL for 2m length and 2mm dia $= \frac{2000}{4} = 500\text{mm}$

ΔL for 1 m length and 1 mm dia $= 1333.33\text{mm}$

$= \frac{1000}{1} = 1000\text{mm}$

65. Which of the following types of light are strongly absorbed by plants?

(a) Violet and orange (b) Blue and red
(c) Indigo and yellow (d) Yellow and violet

Ans. (b) : Photosynthesis occurs between wavelengths of about 400 nm and 750 nm. Red and blue colour wavelength is categories in this wavelength. So plants absorb these colours. Photosynthesis does not occur in the infra-red or in ultraviolet light.

66. Which of the following types is used by computed tomography employed for visualisation of the internal structure of human body?

(a) X-rays (b) Sound waves
(c) Magnetic resonance (d) Radioisotopes

Ans. (a) : Computed tomography is a diagnostic imaging procedure that uses X-rays to build cross-sectional images ("slices") of the body.
CT is based on the fundamental principle that the density of the tissue passed by the X-ray beam can be measured from the calculation of the attenuation of X-ray intensity. Using this principle, CT allows the reconstruction of the density of the body.

67. Which one of the following pairs is not correctly matched?

(a) William Dickson : Motion Picture film
(b) Charles Babbage : Programmable computer
(c) Nicholas Stern : Construction technology
(d) Brian Greene : String theory

Ans. (c) : Nicholas Stern is a British economist and academic.

68. Which one of the following types of glass can cut off ultraviolet rays?

(a) Soda glass (b) Pyrex glass
(c) Jena glass (d) Crookes glass

Ans. (d) : Crookes glass can intercept ultraviolet rays. Ultraviolet rays were discovered by Ritter. The wavelength of these waves ranges from 10^{-8} m to 10^{-7} m. These waves are generated by sunlight, electric discharge, vacuum spark etc.

69. Assertion (A) : A jet aircraft moving at Mach Number equal to 1 travels faster at an altitude of 15 km than while moving at Mach Number equal to 1 near the sea level.
Reason (R) : The velocity of sound depends on the temperature of the surrounding medium.

Ans. (d) : Mach number, a useful quantity in aerodynamics, is the ratio of air speed to the local speed of sound. The speed of sound varies with temperature. Since temperature and sound velocity normally decrease with increasing altitude, sound is refracted upward. Mach number is a function of temperature at altitude. With decrease in sound velocity, Mach number increases.

CHEMISTRY

70. Which one of the following does not contain silver?

(a) Horn silver
(b) German silver
(c) Ruby silver
(d) Lunar caustic

Ans. (b) : German Silver is an alloy of copper, zinc, and nickel, sometimes also containing lead and tin. It does not contain silver metal and is extensively used due to its hardness and resistance to corrosion.

71. Which one among the following is called philosopher's wool?

(a) Zinc bromide
(b) Zinc nitrate
(c) Zinc oxide
(d) Zinc chloride

Ans. (c) : Philosophers wool is a oxide of zinc, a white powder used as a pigment cosmetics glass inks and in zinc ointment.

72. Which one of the following non-metals is not a poor conductor of electricity?

(a) Sulphur
(b) Selenium
(c) Bromine
(d) Phosphorus

Ans. (b) : Sulphur is the best electrical insulating material known, with a resistivity of about 2×1023 μΩ-cm. The reason for this large resistivity is probably the electron traps produced by thermal breaking of S8 rings. Electrical conductivity of sulphur is 5.0×10^{-14} S m^{-1}. Electrical conductivity of selenium is 8×10^6 S m^{-1}. Electrical conductivity of bromine is 1.0×10^{-10} S m^{-1}. Electrical conductivity of phosphorous 1.0×10^{-9} S m^{-1}. Thus from this data it is clear that electrical conductivity of selenium is maximum.

73. Which one of the following is another name of RDX?

(a) Cyanohydrin
(b) Dextran
(c) Cyclohexane
(d) Cyclonite

Ans. (d) : RDX, an initialism for Research Department Explosive, is an explosive nitroamine widely used in military and industrial applications. It is also known as cyclonite. Its chemical name is cyclotrimethylene trinitramine.